Hybrid AI-Based Oral Cancer Detection and Risk Assessment

GitHub README / Project Documentation

Project Overview

This project aims to support oral cancer analysis using Machine Learning and Deep Learning techniques. It consists of two modules: image-based cancer classification using a Convolutional Neural Network (CNN) and questionnaire-based risk assessment using a Random Forest Classifier.

The project is implemented using Python in Google Colab.

Objectives

- Classify oral cavity images into cancer and non-cancer categories.
- Analyse patient risk factors using questionnaire data.
- Classify questionnaire-based risk into Low, Medium, and High levels.
- Explore AI-based methods to support early oral cancer risk assessment.

Modules

1. Image-Based Classification

- Model: MobileNetV2 (CNN)
- Input: Oral cavity images
- Image resizing: 224 × 224 pixels
- Preprocessing: Normalization and data augmentation.
- Output: Cancer Detected / No Cancer.

2. Questionnaire-Based Risk Assessment

- Model: Random Forest Classifier
- Input: Patient questionnaire data.
- Features include age, gender, tobacco use, alcohol consumption, HPV infection, betel quid use, oral hygiene, family history, oral lesions, and other risk factors.
- Risk classification: Level 1 – Low Risk; Level 2 – Medium Risk; Level 3 – High Risk.

Technologies Used

- Python
- Google Colab
- TensorFlow / Keras
- MobileNetV2
- Random Forest Classifier
- Scikit-learn
- Pandas
- NumPy

- Matplotlib
- Seaborn
- OpenCV

Project Workflow

- Collect oral cavity image and questionnaire datasets.
- Preprocess and resize images.
- Train the MobileNetV2 image classification model.
- Evaluate image classification performance.
- Preprocess questionnaire data.
- Calculate risk scores based on patient risk factors.
- Train the Random Forest classification model.
- Predict questionnaire-based risk levels.
- Evaluate model performance using accuracy and confusion matrices.

Repository Structure

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Oral-Cancer-Detection/
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■■■ phase1_img_classfication.ipynb
■■■ phase2.ipynb
■■■ README.md
■■■ requirements.txt
```

Implementation

The project backend is developed using Python in Google Colab. Both models are trained and evaluated independently using their respective datasets.

Future Enhancements

- Integrate both models into a unified prediction pipeline.
- Implement Grad-CAM for visual explanation of image predictions.
- Implement SHAP for explaining questionnaire-based predictions.
- Develop a Flask-based backend.
- Create a user-friendly web application.
- Add doctor recommendation functionality.

Disclaimer

This project is developed for academic and educational purposes. It is a research prototype and is not intended to replace professional medical diagnosis or consultation.

Author

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